

CMPSC463 Project 1

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Description:

The goal of this project was to use the PulseDB dataset, a dataset containing time series data that measured ABP signals. The task was to extract data from this dataset's ABP signal time series by using recursive algorithms in order to separate and cluster the data and extract information from it.

Installation & Usage:

Simply use the given .ipynb file in Google Colab in order to be able to begin using the code given. Note that you must download a segment from the PulseDB dataset and put it into your google drive in order to run the code.

Structure of Code:

The code is all in the singular .ipynb file, and all the functions are written in order of use by the code that reads the given PulseDB dataset segment file.

Algorithms:

`compute_distance_matrix` uses the `fastdtw` python package in order to find the most dissimilar time series for use in clustering, and `divide_and_conquer_clusters` then splits the two clusters into more clusters

`lb_keogh` is a function used by the `closest_pair_dtw` function to help prune the results by pruning the `closest_pair_dtw` results

`KanadeAlgo` is a simple $O(n)$ algorithm that calculates the max value of the possible subarrays by keeping track of the maximum value it runs into.

Execution Results:

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Closest pair: (1, 2), DTW distance = 73.185
Closest pair: (0, 1), DTW distance = 82.504
Closest pair: (1, 2), DTW distance = 96.144
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981: 9468.661
982: 9465.806
983: 9457.751
984: 9444.459

985: 9426.782
986: 9400.346
987: 9371.875
988: 9346.907
989: 9314.872
990: 9285.941
991: 9256.954
992: 9222.159
993: 9198.240
994: 9170.858
995: 9140.718
996: 9123.619
997: 9104.725
998: 9088.231
999: 9080.364
1000: 9079.114

Discussion of Execution Results:

Closest Pairs seems to return none at some points, this is likely to be a problem with clustering as perhaps it creates empty clusters for some odd reason.

There are 1000 results for the maximum sums given by the Kanade Algorithm, unfortunately I don't know enough medical to inference anything meaningful.

Conclusions:

Admittedly this project is hampered by a lack of knowledge on the subject, further research and fine tuning of code is required.